

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1662

Solar 138 kV
July 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1662. The Interconnection Customer proposes a 185.7 MW battery energy storage facility interconnecting at the Solar 138 kV substation in the Virginia area. The request was received on February 04, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 496 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,212,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Stepdown-Alpine 138 kV	Mira Vista-Yarnen 345 kV	143.0	310	7.2
Hardy Reit-Stepdown 138 kV	Yarnen Substation-Mira Vista 345 kV	120.9	473	8.7
Mitchell Dam-Syl Laskin 138 kV	Turner Rd Substation-Tyler 138 kV	119.6	228	5.0
Chico A Substation-York Canyon 138 kV	Yarnen-Mira Vista 345 kV	117.8	417	5.7
Hardy Reit-Battlefield 138 kV	Mira Vista-Yarnen 345 kV Line	117.2	447	13.6
Northern Neck-Chico A 138 kV	Yarnen-Mira Vista 345 kV	116.1	448	5.7
Solar-Coppell 138 kV	Coppell-Stoughton 138 kV	108.1	229	62.7
Thrasher-Wolcott 138 kV L	Mira Vista-Yarnen 345 kV	106.1	378	12.7
Calpine King Energy-Wolcott 138 kV	Mira Vista-Yarnen Substation 345 kV	105.2	398	12.7
Kenneth C Coleman-Hammerhill 138 kV	Yarnen-Mira Vista 345 kV	104.8	184	5.2
Cochise Substation-Bobville 138 kV	Mccartney-Yarnen 345 kV	104.3	317	7.0
Roanoke Rapids-Fyffe 138 kV Ckt 1	Yarnen-Mccartney 345 kV line	104.2	539	13.2
Solar-Abilene South 138 kV	Coppell-Solar 138 kV Line	103.7	215	100.0
Coppell-Mcindoes 138 kV	Coppell-Mcindoes 138 kV	103.6	283	34.1
Tillman-Carroll 138 kV	Mira Vista-Yarnen 345 kV	103.3	402	16.2
Calpine King Energy 345/138kV Transformer Bank	Yarnen-Mira Vista 345 kV	103.2	386	12.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
No Mans Land-Ne-Hobbs Substation 138 kV	Mira Vista-Yrnen 345 kV	101.9	442	13.8
Roanoke Rapids-Fyffe 138 kV #2	Mccartney-Yarnen 345 kV	101.9	551	13.2
Pine Bluff W-Lake Kingman 138 kV line	Yarnen-Mira Vista 345 kV line	100.9	258	17.7
Spring Street-Chipola 138 kV	Skaggs Island Junction 2-Mitchell Dam 345 kV line	100.7	234	7.0
John R Kelly-Scott Paper Co 138 kV	Pana Substation-Roney 138 kV Line	100.6	251	8.9
Wheelabrator Portsmouth-Reeves Ave 138 kV	Mickens-Mira Vista 345 kV L	100.3	245	5.3
Danc Lfgte Facility-Bonny Brook 138 kV	Mira Vista-Yarnen 345 kV Line	100.1	366	9.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stepdown-Alpine 138 kV	Rebuild line with higher-capacity conductor	\$5,997,000	5% (\$315,000)
Hardy Reit-Stepdown 138 kV	Rebuild line with higher-capacity conductor	\$11,987,000	5% (\$617,000)
Mitchell Dam-Syl Laskin 138 kV	Rebuild line with higher-capacity conductor	\$1,605,000	11% (\$184,000)
Chico A Substation-York Canyon 138 kV	Rebuild line with higher-capacity conductor	\$11,279,000	7% (\$796,000)
Hardy Reit-Battlefield 138 kV	Rebuild line with higher-capacity conductor	\$25,496,000	6% (\$1,439,000)
Chio A-Northern Neck 138 kV	Rebuild line with higher-capacity conductor	\$2,904,000	7% (\$205,000)
Solar-Coppell 138 kV	Reconductor line	\$1,140,000	100% (\$1,140,000)
Thrasher-Wolcott 138 kV L	Reconductor line	\$24,528,000	11% (\$2,741,000)
Calpine King Energy-Wolcott 138 kV	Reconductor line	\$7,213,000	11% (\$806,000)
Kenneth C Coleman-Hammerhill 138 kV	Reconductor line	\$4,859,000	10% (\$465,000)
Cochise Substation-Bobville 138 kV	Reconductor line	\$3,803,000	5% (\$182,000)
Fyffe-Roanoke Rapids 138 kV Line 1	Reconductor line	\$13,498,000	6% (\$801,000)
Solar-Abilene S 138 kV	Reconductor line	\$1,826,000	100% (\$1,826,000)
Coppell-Mcindoes 138 kV	Reconductor line	\$3,240,000	60% (\$1,926,000)
Tillman-Carroll 138 kV	Reconductor line	\$1,433,000	9% (\$127,000)
Calpine King Energy 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$20,885,000	11% (\$2,335,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
No Mans Land-Ne-Hobbs 138 kV	Reconductor line	\$1,303,000	5% (\$60,000)
Roanoke Rapids-Fyffe 138 kV #2	Reconductor line	\$12,804,000	6% (\$760,000)
Pine Bluff W-Lake Kingman Substation 138 kV	Reconductor line	\$939,000	11% (\$100,000)
Spring St-Chipola 138 kV	Reconductor line	\$3,308,000	3% (\$107,000)
John R Kelly-Scott Paper Co 138 kV	Reconductor line	\$902,000	53% (\$474,000)
Wheelabrator Portsmouth-Reeves Ave 138 kV	Reconductor line	\$1,162,000	55% (\$642,000)
Danc Lfgte Facility-Bonny Brook 138 kV	Reconductor line	\$2,603,000	6% (\$164,000)
	Total allocated to project		\$18,212,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.